

Portfolio

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Semantic Segmentation

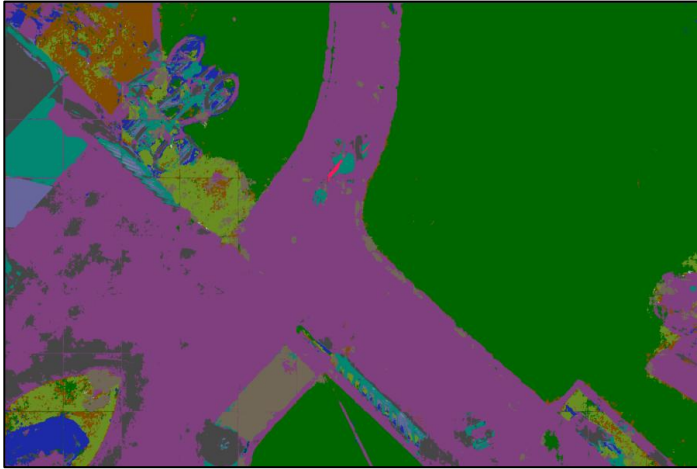


Figure 1. Segmented image example.

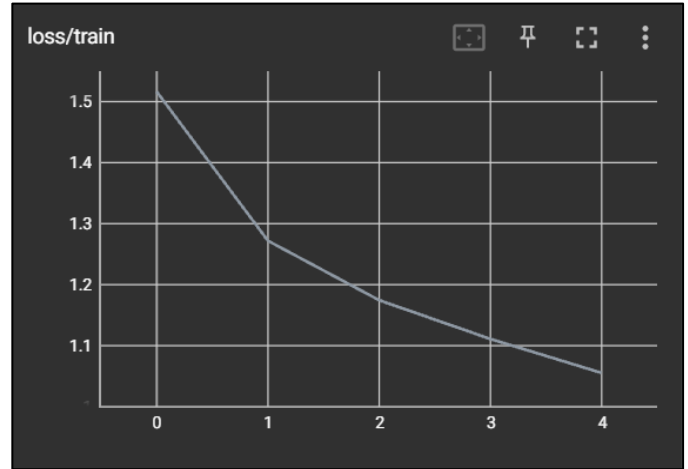


Figure 2. Dice loss vs. training epoch.

- Developed semantic segmentation program with PyTorch to classify each pixel of images collected by aerial robots
- Defined custom U-Net model and trained on the Semantic Drone Dataset by the Graz University of Technology, yielding over 75% validation accuracy in only 4 training epochs
- Patched 4000 x 6000-pixel images into 500 x 500-pixel patches to make training and application of the model computationally feasible, even for limited GPUs
- **Project repository available [here](#)**

12S 3A Drone BEC

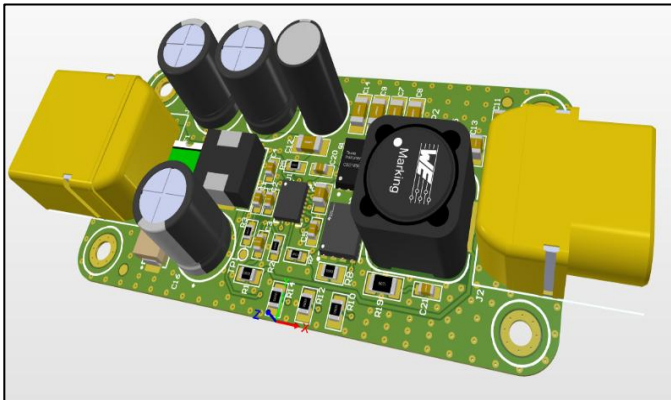


Figure 3. BEC Altium render.

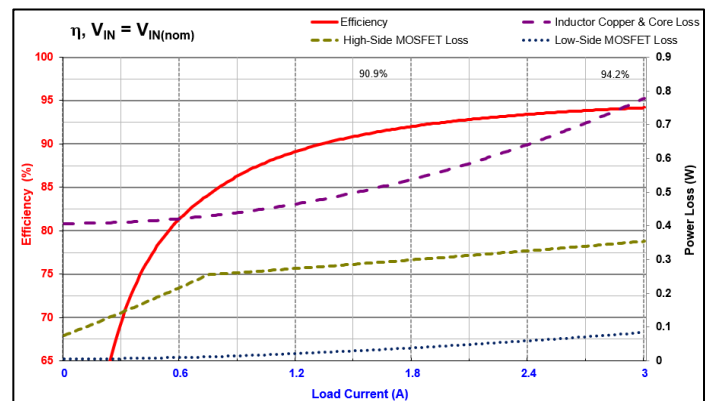


Figure 4. BEC efficiency for 12V output.

- Designed BEC to convert 3S to 12S LiPo inputs to 5.1V, 8V, and 12V 3A outputs on RC drones
- Modelled buck controller, inductor, and MOSFET combinations to yield 94% worst-case efficiency for a 12V output and 87% for a 5.1V output at maximum load
- Implemented overcurrent protection, spike protection, undervoltage lockout, and soft start
- Configured Type-III compensator with 60° to 65° phase margin for control-loop stability
- **Technical whitepaper available [here](#)**

Payload Sensor Module

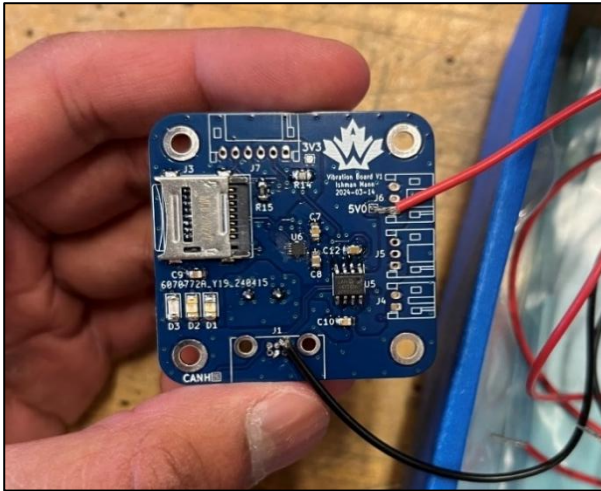


Figure 5. PSM prototype.

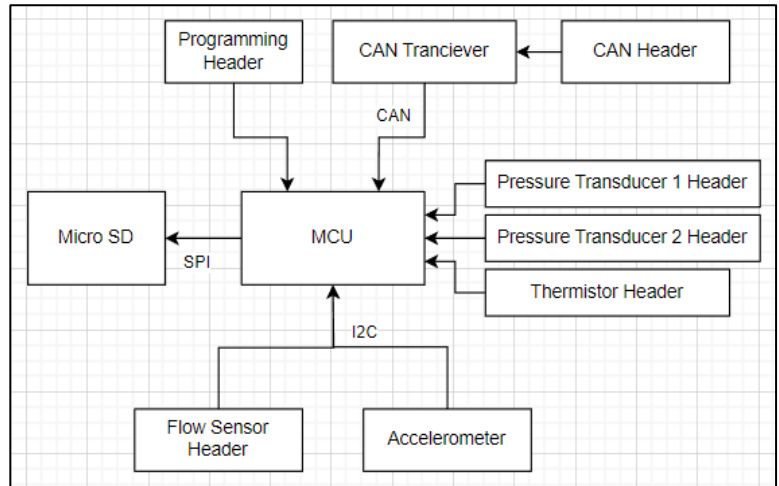


Figure 6. PSM system architecture.

- Designed module to facilitate mid-flight blood-transfusion and mechanical-vibration experiments in the payload of Waterloo Rocketry's 2024 rocket using a suite of sensors
- Sported a 16g 3.2kHz accelerometer, 200Hz flow sensor, thermistor, and pressure transducers
- Took care in considering temperature effects and signal integrity to ensure sampling accuracy
- Routed paths for micro-SD data storage, CAN bus communication, and PIC18 MCU debugging
- **Technical whitepaper available [here](#)**

ELRS Gemini Xrossband TX

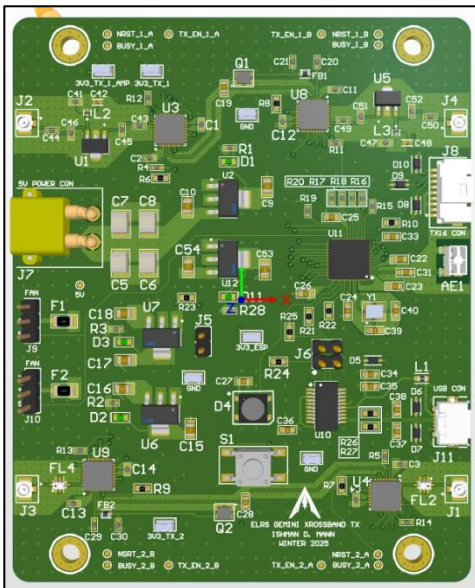


Figure 7. TX module Altium render.

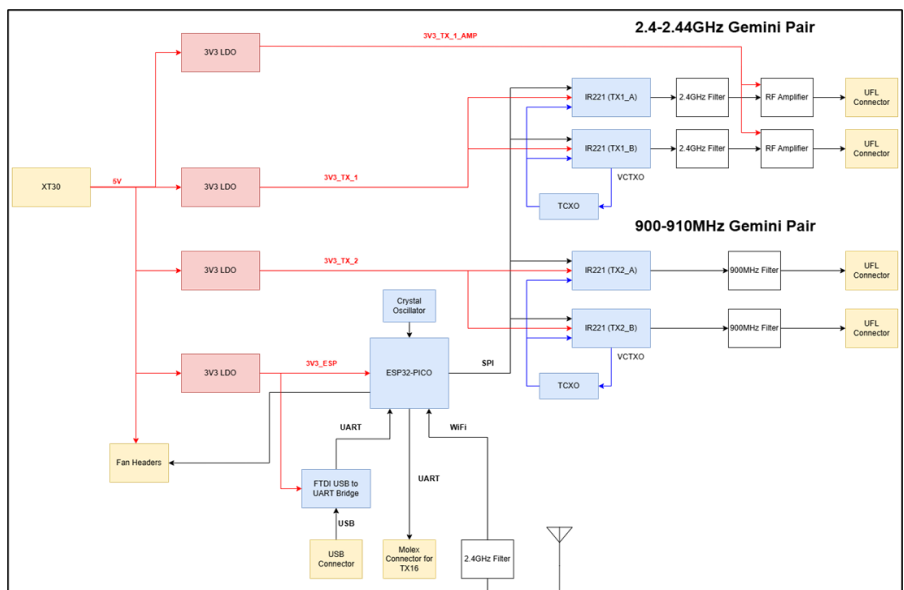


Figure 8. TX module system architecture.

- Designed ELRS Gemini Xrossband TX module for dual-band communication with RC drones
- Supported independent and simultaneous 27dBm transmission of low-latency 2.4-2.44Ghz and long-range 900-910MHz LoRa signals using pairs of LR221 chips synchronized by TCXO
- Enabled USB/Wi-Fi flashing and TX16 UART control of central ESP32-PICO microcontroller
- Defined 6-layer stackup with impedance-controlled microstrips and coplanar waveguides
- **Technical whitepaper available [here](#)**